El. motor Load classification based on the angular speed

The motor drives a load which has a certain characteristics torque-speed requirement.

In general, the general law of the load torque is:

$$T_L = T_0 + (T_n - T_0)(\frac{\omega}{\omega_n})^q \quad \text{.....(1)}$$

where, q may be an integer (…-2,-1,0,1,2,...), ω is the angular speed of the load ω_n –rated angular speed of the load machine, T_n rated load in N.m. T_0 initial load in N.m.

According to the general law, most loads faced in the industry can be classified into four groups:

1- Group (1)-Constant-power loads:

They have non linear (falling) characteristic and consists of machines for winding material under tension. This group includes, for example veneer (قشره) cutting machines (lathes) and machine tools. q= -1, if we neglect the initial conditions, assume T_0 =0 , the speed torque relation is:

$$T_L = (T_n)(\frac{\omega}{\omega_n})^{-1} = \frac{T_n \omega_n}{\omega} = \frac{k}{\omega} \quad \text{.....(2)} \qquad P == T_L \omega = \frac{k}{\omega}\omega = k \quad \text{.....(3)}$$

Figures of speed-torque and speed-power characteristics are shown in the table.

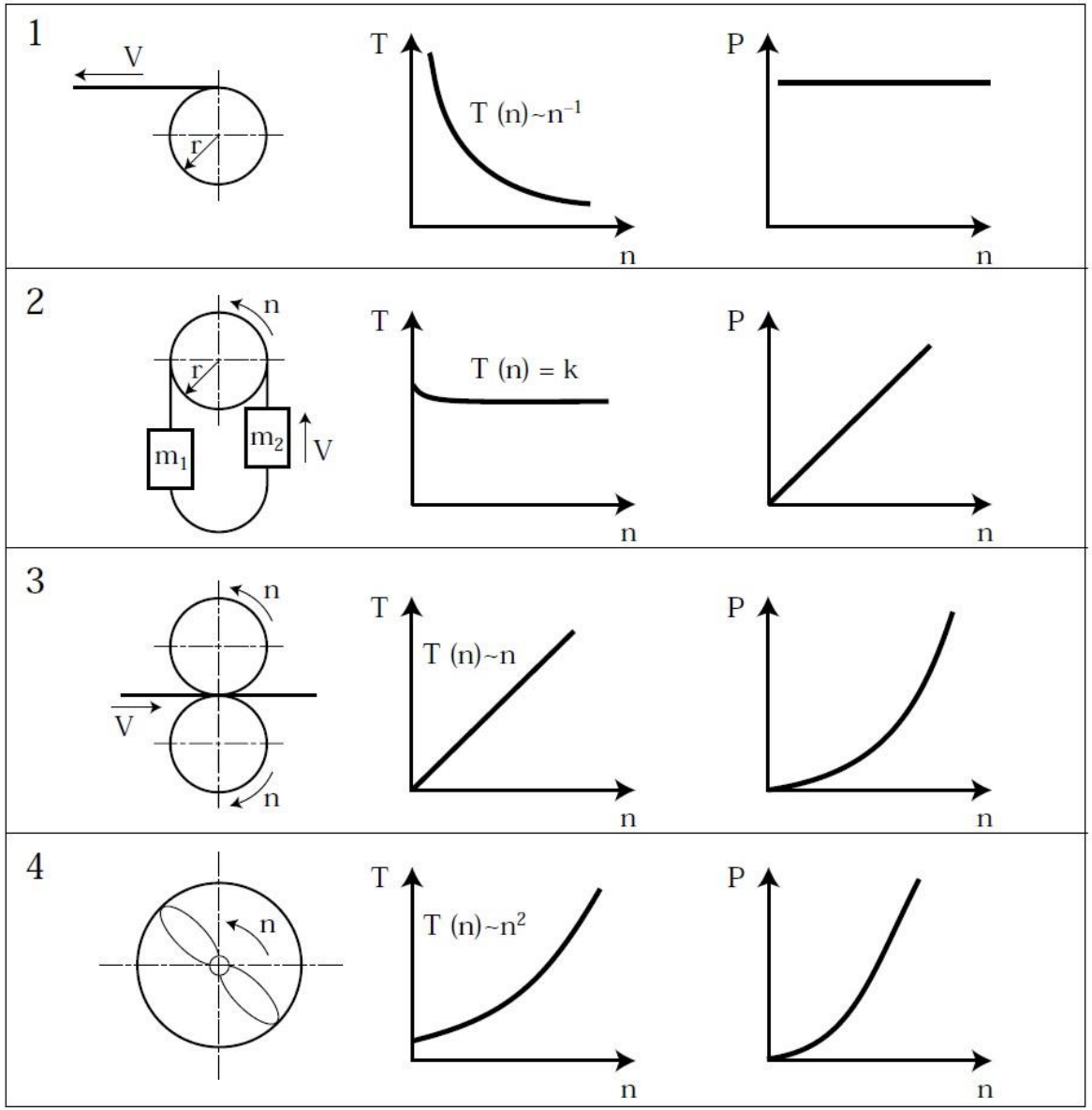
2- Group (2) –Constant torque loads:

q= 0, if we neglect the initial conditions, assume T_0 =0 , the speed torque relation is:

$$T_L = (T_n)(\frac{\omega}{\omega_n})^0 = T_n \quad \text{.....(4)} \qquad P = T_L \omega = T_n \omega = k\omega \quad \text{.....(5)}$$

This means that the load value doesn't depend on the speed. The power is dependent on the load speed. Figures of torque and power characteristics versus speed are shown in the table.

examples: elevators, escalators, conveyor belt, cranes, positive displacement pumps, paper mill drive, piston compressors as well as machine tools.



Group (3): Machineries with linear incremental characteristics. They have linear rising characteristic and consists of machines such as hydraulic pumps, rollers (اسطوانات الدرفلة السفلية), smoothing machines (الات السنفرة), and other processing machines

q= 1, if we neglect the initial conditions, assume T_0 =0 , the speed torque relation is:

$$T_L = (T_n)(\frac{\omega}{\omega_n})^1 = \frac{T_n\omega}{\omega_n} = k\omega \text{.....(6)} \qquad P = T_L\omega = \frac{T_n\omega}{\omega_n}\omega = k\omega^2 \text{.....(7)}$$

Speed-torque and speed-power characteristics are shown in the table

Group (4)-Quadratic torque :

fan loads (non linear rising or parabolic characteristic) - comprises machines operating by centrifugal force, such as centrifuges, centrifugal pumps , fans and rotary vane compressors .

q= 2, if we neglect the initial conditions, assume T_0 =0 , the speed-torque and speed-power relation are:

$$T_L = T_n\left(\frac{\omega}{\omega_n}\right)^2 = \frac{T_n\omega^2}{\omega_n^2} = k\omega^2 \text{ ..... (8)} \qquad P = T_L\omega = \frac{T_n\omega^2}{\omega_n^2}\omega = k\omega^3 \text{.....(9)}$$

characteristics of this group are shown in the table

Load classification based on the motion direction

- **Various motor loads can be broadly classified as:**

1. Active loads: they have the potential to drive the motor under equilibrium condition such as torques due to gravitational force, tension, compression and torsion , undergone by an elastic body. They retain their sign when the direction of the drive motion is changed.
2. Passive loads: load torques which always oppose the motion and change their sign on the reversal of motion such as torques due to friction, windage ,cutting etc.

Components of the load torque (Dependence of load torque on various factors)

- Load torque T can be divided into three components:

1. **Friction torque (T_f)** **2.windage torque (T_w)** **3. work required to be the useful mechanical work (T_L)**

Friction torque: Friction will be presented at the motor shaft and also in various parts of the load. T_f -Is the equivalent value of various friction torques referred to the motor shaft. Variation of friction torque with speed is shown in figure 2.6a

T_f - value at stand still is much higher than its value slightly above zero speed.

Friction at zero is called stiction (التصاق)or static friction.

In order to start the drive, the motor torque should at least exceed the stiction.

Friction torque can be resolved into three components as shown below(2.6b)

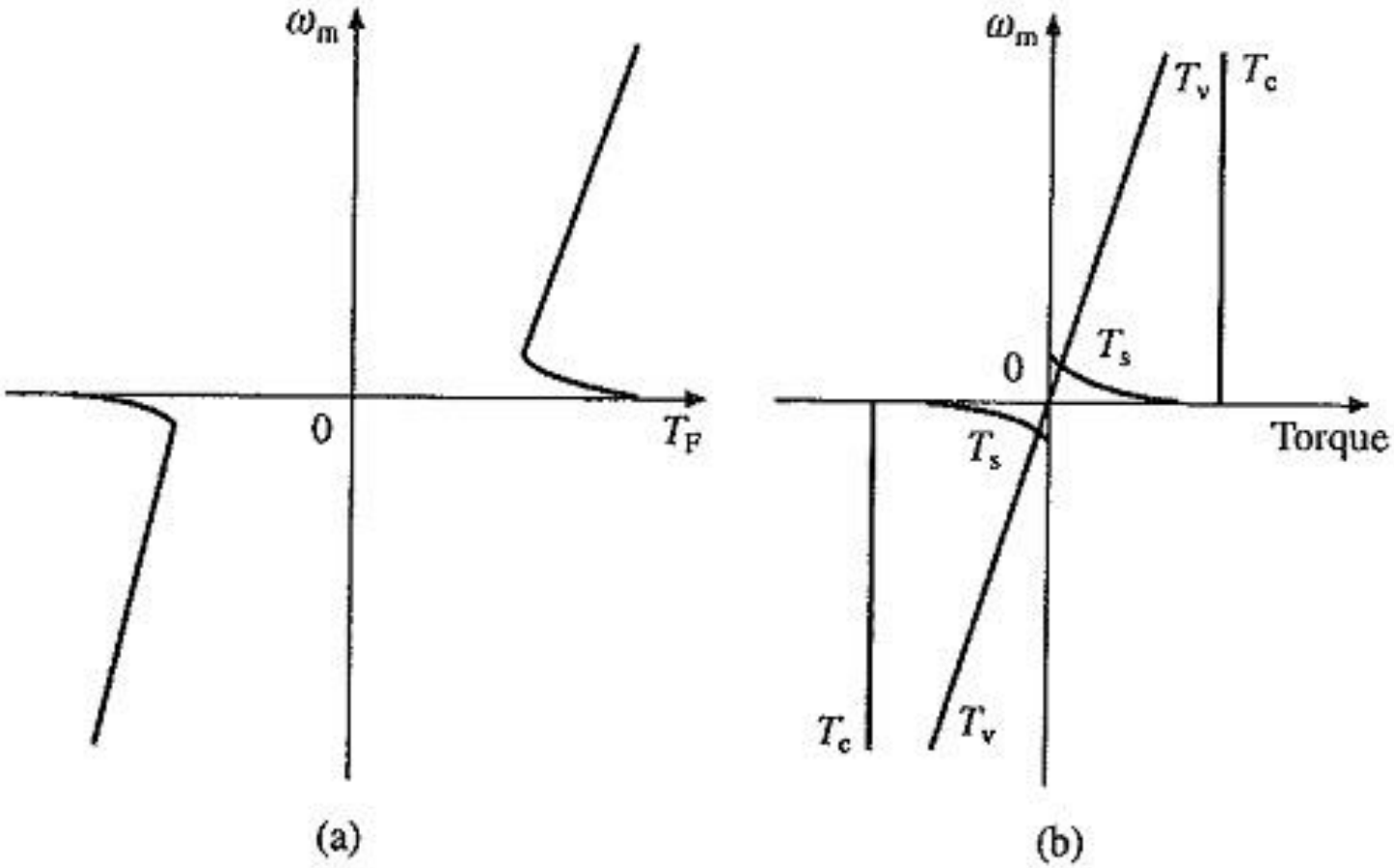


Fig. 2.6 Friction torque and its components

Component T_v which varies linearly with speed is called viscous friction and is given by $T_v = b\omega_m$

Where b is the viscous friction coefficient

Another component Tc is independent of speed known as Coulomb friction.

The third component T_s accounts for additional torque present at a standstill. Since T_s is present at a standstill, it is not taken into account in the dynamic analysis.

$$T_f = b\omega_m + Tc + T_s = b\omega_m + Tc$$

2. Windage torque (T_w), when the motor rotates, wind generates a torque opposing the motion, this is known as windage torque.

Windage torque is proportional to the speed square.

$T_w = C\omega^2$, where C- proportional constant for windage torque.

3. Torque required to do the useful mechanical work (T_L)

Nature of this torque depends on particular application

a. It may be constant and independent of speed
b. It may be some function of speed
c. It may depend on the position or path followed by the load
d. It may be time-invariant or time-variant
e. It may vary cyclically and its nature may also change with the load's mode of operation.

From the above discussion, for the finite speed, load torque is

$T_l = T_L + b\omega_m + Tc + C\omega^2$

In many applications, ($Tc + C\omega^2$) is very small compared to($B\omega_m$) and negligible compared to (T_L)

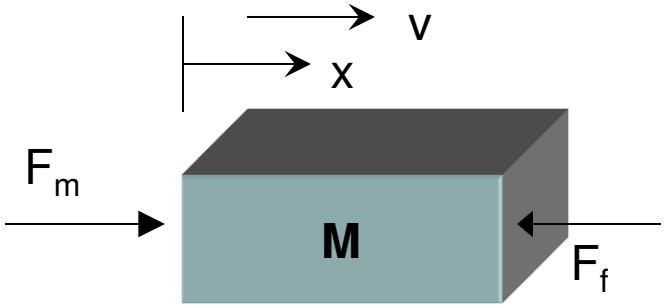
To simplify the analysis, the term ($Tc + C\omega^2$)is accounted by updating the value of viscous friction coefficient b. With this approximation,

$$T_l = T_L + B\omega_m$$

For medium and high power loads, the second component is small compared to the first one and can be omitted, so

$$T_l = T_L$$

Torque equation for translational movement



Newton's law

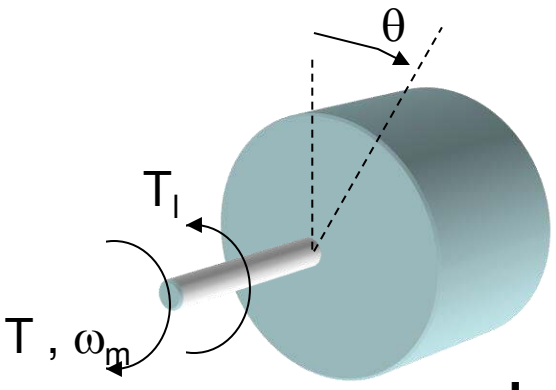
$$F_m - F_f = \frac{d(Mv)}{dt}$$

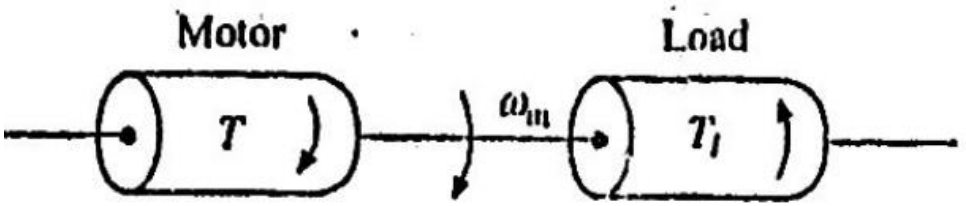
Linear motion for constant mass (M),

$$F_m - F_f = M\frac{d(v)}{dt} = M\frac{d^2x}{dt^2} = Ma$$

- First order differential equation for speed
- Second order differential equation for displacement

Torque equation for rotational motion





- Normally is the case for electrical drives

By neglecting the friction component, torque equation can be written as:

$$T - T_L = \frac{d\left(J\omega_m\right)}{dt} = J\frac{d\omega_m}{dt} + \omega_m\frac{dJ}{dt}$$

With constant moment of inertia (J),

$$T - T_L = J\frac{d\omega_m}{dt} = J\frac{d^2\theta}{dt^2}$$

- First order differential equation for angular frequency (or velocity)
- Second order differential equation for angle (or position)

Motor torque and power equations

The motor torque should overcome the static load and dynamic load

$$T = T_{ref} + J_{ref}\frac{d\omega_m}{dt}$$

When the motor shaft torque is equal to the torque exerted by the load ($T_L = T_{st} = T_{\Sigma} = T_{ref}$), the motor is stationary. In such cases the torque and the speed are constant. The motor works on steady state (constant speed), the dynamic component becomes zero, So, the stationary state occurs when motor and machine torque are identical. The power. voltage and speed given in the nameplate of the motor are its rated values, i.e for continuous steady-state operation, The motor power for continuous steady-state operation is calculated as:

$$P = kT_{ref}\omega_m$$

where, k= (1.1-1.3) - is a safety factor and it depends upon the motor power

T_L ,T_{st},T_{ref} - load, static or referred (equivalent) torque

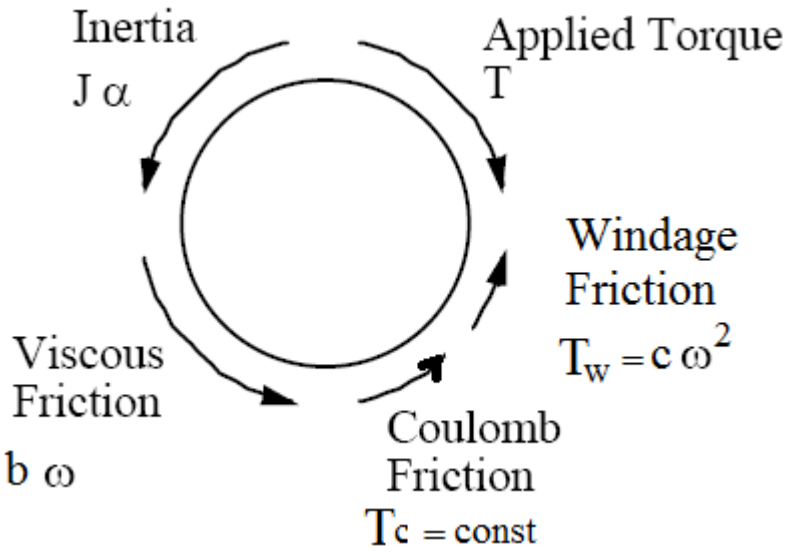
Motor torque

According to the motion equation, The motor can have acceleration, deceleration or steady behavior

$T_m > T_L$, $d\omega/dt > 0$ متسارع عندما

$T_m < T_L$, $d\omega/dt < 0$ متباطئ عندما

$T_m = T_L$, $d\omega/dt = 0$ مستقر عندما



The motion equation without neglecting small components is:

$T = K_t I = T_L + J\, d\omega/dt + B\,\omega$

where : B- updated viscous friction coefficient

B ω – viscous friction torque + Coulomb friction + windage friction

Load Equalisation (موازنة التحميل)

- In some drive applications, load torque fluctuates widely within short intervals of time.
- For example, in pressing machines a large torque of short duration is required during pressing operation, otherwise, the torque is nearly zero.
- Other examples are electric hammer, steel rolling, mills and reciprocating pumps. (من الأمثلة الأخرى المطرقة الكهربائية ودرفلة الفولاذ والمطاحن والمضخات الترددية)
- For such driving, if the motor is required to supply peak torque demanded by load, the first motor rating has to be high. Secondly, the motor will draw a pulsed current from the supply.
- When the amplitude of pulsed current forms an appreciable proportion of the supply line capacity. It gives rise to line voltage fluctuations, which adversely affect other loads connected to the line.
- In some applications, peak load demanded may form a major proportion(نسبة كبيرة) of the source capacity itself, as in blooming mills(درافل النورات), then load fluctuations may also adversely affect the stability of the source.
- These problems of fluctuating loads are overcome by mounting a flywheel on the motor shaft in non-reversible drive ($J=J_m+J_{flw}$).

- Motor speed- torque characteristic is made drooping (characteristic AC shown in the below figure.
- Alternatively, in closed-loop current control, torque

Is prevented from exceeding a permissible value (characteristic ABC in the figure)

- Driving high load period, load torque will be much larger compared to the motor torque. Deceleration occurs producing a large dynamic torque component ($J\frac{d\omega}{dt}$).

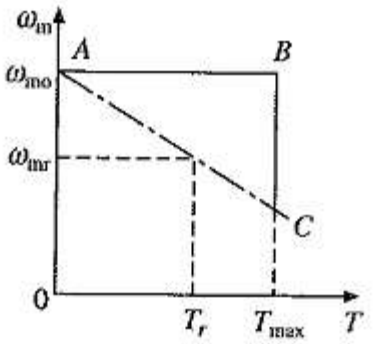


Fig. 2.10 Shapes of motor speed torque curves for fluctuating roads

- Dynamic torque and motor torque together are able to produce torque required by the load ($T = T_l + J\frac{d\omega}{dt}$), where $\frac{d\omega}{dt}$<0. Because of deceleration, the motor speed falls.
- During the light load period, the motor torque exceeds the load torque causing acceleration. Speed is brought back to its original value before the next high load period.

- Variation of motor and load torques, and speed for a periodic load and for a drooping motor speed torque are shown in the figure below.
- From this figure, its shown that peak torque required from the motor has much smaller value than the peak load torque.
- Hence, a motor with much smaller rating than peak load can be used and peak current drawn by the motor from the source is reduced by a large amount.
- Fluctuations in motor torque and speed are also reduced. Since, the power drawn from the source fluctuates very little, this is called Load equalisation.

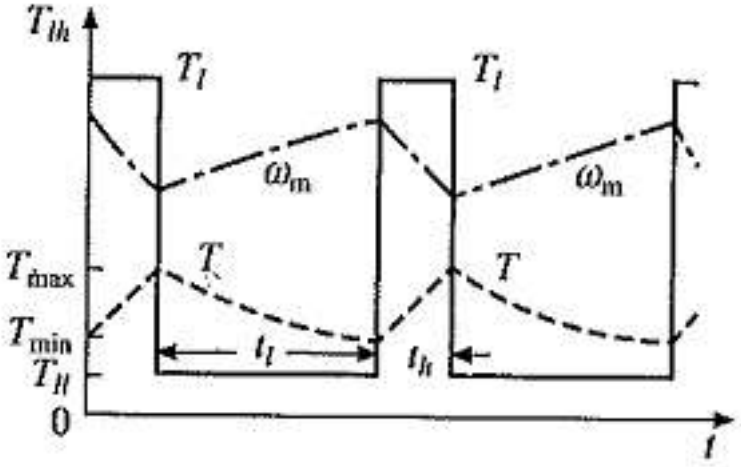


Fig. 2.11

- In variable speed and reversible drive, a flywheel cannot be mounted on the motor shaft as will increase the transient time of the drive by a large amount.
- If the motor is fed from motor-generator set (Ward-Leonard drive), then flywheel can be mounted on the shaft of the motor-generator set. This arrangement equalizes the load on the source but not the load on the motor. Consequently, a motor capable of supplying peak-load torque is required(مطلوب محرك قادر على توفير عزم الدوران عند ذروة الحمل).